Machine Learning Methods for Surrogate Observables in Lattice Quantum Chromodynamics

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ABSTRACT
Lattice Quantum Chromodynamics requires computationally expensive calculations to obtain physical observables describing strong-force interaction dynamics, which typically require large-scale supercomputing. The high computational cost arises primarily from the need to parallelise an immense number of calculations. As an alternative approach to reduce this computational expense, this project utilises machine learning models trained on correlator data to learn a mapping from computationally cheaper operators to more expensive operators via regression. This project compared a bias correction method with a more widely tested ratio-based correction method for machine learning outputs and fitted correlator data using standard fitting techniques. The data revealed that a bias correction method yielded results comparable to the benchmark, even outperforming the ratio method for some models.

Key words: Lattice Quantum Field Theory – Machine Learning – Lattice QCD

1 INTRODUCTION

The most significant limitation of lattice QCD is the matrix inversions for quarks. To invert a spacetime grid that has 48^3 spatial sites and 96 temporal sites (which bears similarity to the grid used by this experiment), we would have a grid of 10,616,832 sites overall, each site has a quark with 12 complex degrees of freedom (4 spin by 3 colour), which leads to over 127 million degrees of freedom. To calculate the Dirac Operator, we invert a rank-120-million vector space, which is used to compute the hadron propagator.

Additionally, mesons with staggered fermions (that utilise Non-Goldstone operators) tend to have higher noise-to-signal ratios, making them less than ideal for computation, as more sampling and ensemble averaging are required compared to their cleaner Goldstone counterparts.

This is a significant computational limitation (especially for lighter mesons, which have far more statistical noise in their signals) that has led to the development of methods to create surrogate variables. One of the most popular of such methods is the Truncated Solver Method (TSM), which often involves computing a large group of inversion matrices at lower precision for most of the process and then comparing a lower precision matrix to the higher precision matrix to find a bias correction factor that is then applied to those matrices (this method is the motivation for the bias correction method utilized in this paper). However, the most significant limitation of this model is that it does not generalise well and requires computing matrices for new data of a different type.

Much like the truncated solver method, a variance-reduction method, such as the ratio method, also aims to create surrogate variables that reduce computational cost. This method assumes a multiplicative relationship between the measured and predicted correlator, which is often the case, hence its popularity. However, its multiplicity is also its caveat: for hadrons with low-momentum operators, the ratio tends to have a denominator that approaches zero, leading to a ratio limit that is higher than needed.

As such, an emerging field that helps address such problems is machine learning. The usage of supervised learning methods to learn a mapping between Goldstone and Non-Goldstone operators on a small subset of the available lattice data is now an area of particular interest as a novel alternative to traditional surrogate variable methods.

However, machine learning predictions are not always stable and can be error-prone, especially when the model itself learns biases or overfits to noisier regions of the data. From this, another primary requirement to ensure we have physically meaningful data is often to compute a correction factor.

This study compares a machine learning bias-correction approach with the widely used Ratio Method, encouraging a sense of progress and collaborative effort in improving estimates.

The study then focuses on obtaining reasonable estimates from the ML model to serve as baselines for fitting the correlator data to obtain values for the data's ground-state amplitude, ground-state effective energy, and a reasonable fit. From which it is revealed that the Machine Learning + bias correction method, alongside the machine learning model, helps attain very effective fits that compare very competitively with the truth, and the Machine Learning + ratio method fits as well. El-Khadra & Vega (2024)

The results demonstrate that machine learning methods combined with techniques like bias correction computation can significantly reduce computational costs in Lattice QCD.

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2 THEORY

2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is a non-Abelian gauge theory that explains the strong force interactions between quarks and gluons (force carriers). Gluon fields help maintain Lagrangian invariance under local transformations, hence why it is a gauge theory.

At the core of QCD lies the SU(3) symmetry group. The SU(3) symmetry group is a group of 3×3 matrices known as the Gell-Mann matrices. To be considered special and unitary, they must satisfy two conditions.

The requirements for being a Unitary matrix (which is a complex analogue of orthogonal matrices) are to satisfy the following equation:

$$U^{\dagger}U = I \tag{1}$$

Where $U^{\dagger}$ is the hermitian adjoint (complex conjugate transposed) of the matrix.

$$det(U) = 1 \tag{2}$$

It is as a direct consequence of these matrices that this gauge theory is non-Abelian (as matrix operations violate commutation properties, unlike its electrodynamic U1 counterpart). These aspects of QCD make it so that there are gluon-gluon interactions that form structures that tend to cause a coupling α_s that is low at high energies and low distances while being higher at lower energies and higher distances (about $\alpha_s \approx 0.12$ which is much more than its QED counterpart $\alpha \approx \frac{1}{137}$

This, in turn, means that standard perturbative methods do not work well in low-energy regimes. Due to the nature of the coupling constant α_s, we cannot calculate the interaction terms using a simple series expansion, as higher-order terms. As a direct consequence of this, the Feynman path integrals that are required to calculate correlation functions are outright impossible to solve analytically. This means a perturbative QCD theory is not feasible at low energies. As such, we need a method that can numerically compute high-accuracy approximations in the low-energy unperturbed states that we often deal with in QCD. Herein lies the motivation for a theory that can reduce the complexity of infinite-dimensional Feynman path integrals over the quark and gluon fields.

2.2 Lattice Quantum Chromodynamics

Lattice QCD provides an accurate approximation to the infinite-dimensional Feynman Path Integrals by discretising spacetime into a 4-dimensional lattice (3 Spatial Dimensions and 1 Temporal dimension, to be consistent with special relativity). The structure of the lattice involves spatial points that often represent quark fields connected by gluon field lines. Each quark field is subject to 4 degrees of freedom from its Dirac Spinor components and 3 degrees of freedom from the colour charge components. This results in a total of 12 degrees of freedom at any given point in a lattice.

By the discretisation of spacetime into a lattice of $L^3 \times T$ lattice where L is the number of lattice spatial extent and T is the lattice temporal extent (for this study $T = 96$), we can now begin to try and model the action of the quark fields (which is essential to the Feynman Path Integral). To do this, the Wilson-Dirac Operator (I will be referring to this as the Dirac Operator or M) for the quark fields is essential, as when we use the Berezin (or Grassmann) integrals to compute the integral of the Quark fields, we get the determinant of the Dirac Operator $det(M)$. We also find that the probability of a gluon configuration (U) is proportional to the determinant of the Dirac Operator of the configuration and its action. Gattringer & Lang (2010)

This means we can now use a special set of methods known as Markov Chain Monte Carlo (MCMC) to find probabilities of configurations occurring and sample accordingly. Once we have the most likely configurations, we can also use the inverse of the Dirac Operator (M^{-1}) quark propagator, which helps define the probability of quark movement between points. The inversions of these Dirac Operators tend to be the most computationally expensive part of Lattice QCD, especially for lower mass hadrons, where it takes longer for the inversion algorithms to converge to accurate values. These quark propagators will be essential to the correlator functions, which will help deduce fundamental properties of the configurations. Xhako et al. (2022)

2.3 Correlators

To measure characteristics of mesons (and other hadrons), the construction of an operator (S) that accurately represents the meson is essential (in this study's case, it would refer to operators that are used to construct D mesons). Then one needs to define the type of correlation. There are two main types of correlations.

Two-point correlations (the primary focus of this study) involve the creation of the meson at a source point, its propagation, and its annihilation at the sink.

Three-point correlations (which will be briefly discussed in the discussion section of this paper) involve the creation of the operator at a source, some form of external change, such as a flowing current, occurs before the annihilation at the sink.

Once the required correlation is chosen, the correlation function ($C(t)$) can be constructed through a process known as Wick Contraction. Gattringer & Lang (2010)

Once we obtain the correlator function $C(t)$, we can then use a process known as spectral decomposition to understand the sum of energy states for the meson, as shown by the formula below:

$$C_H(t) = \sum_{i=0}^{N_{\text{exp}}} (|d_i^{H,n}|^2((e^{-E_i^{H,n}t} + e^{-E_i^{H,n}(N_t - t)})) \tag{3}$$

Where H refers to the hadron measured. N_{exp} denotes the number of exponentials you will fit. The $d_i^{H,n}$ is the amplitude for the ith state. $E_i^{H,n}$ will refer to the energy at the ith state. The two periodic exponential terms show the torus-like shape formed by the lattice, allowing backward propagation around the boundary. There won't be much emphasis on the 3-point correlator functions in the theory. Still, their equations are similar, with the inclusion of terms to account for an external current factor, and they utilise a double sum for insertion states (going to interaction points) and outgoing states (going from interaction points).

When fitting, we can approximate this into a fit of 2 states that follows the following equation:

$$C_D(t) \approx a_o e^{-E_0 t} + a_1 e^{-E_1 t} \tag{4}$$

Where a_0 and a_1 are the respective ground state and first excited state amplitudes. E_0 and E_1 are the respective ground and first excited state energies (which can be used to calculate the effective masses). El-Khadra & Vega (2024)

We can extract useful information such as the effective mass of a correlator using the following equation:

$$m_{eff} = ln\left(\frac{C(t)}{C(t+1)}\right) \quad (5)$$

2.4 Correlator Noise

In Lattice QCD, 2-point correlator functions exhibit exponential decay with increasing time, while the associated statistical noise tends to decrease more slowly. As a direct consequence of this behaviour, for large values of time, the noise-to-signal ratio grows larger towards the midpoint of the temporal lattice. This effect limits our ability to extract important information from the correlator functions effectively. A large noise-to-signal ratio is even more prevalent in noisier correlators, such as the non-Goldstone meson operators, which typically include larger contributions from excited and noisier states. This becomes even more important towards times closer to the point of inflection for the lattices as the signal of the correlators tends to drop the most at this area meaning most of the calculated signal is noisy. Therefore, it is often required that we use large ensembles of data and higher-precision inversions of the Dirac operator. This tends to be computationally expensive, which is why it is usually cheaper to generate Goldstone operators, which are more centred on ground-state contributions and generally low-mass. It is also why it is essential to examine techniques that help us build variables and data that are genuinely effective estimates, thereby reducing computational cost.

2.5 Surrogate Observables

Surrogate observables (also referred to as surrogate variables) in lattice QCD are approximate observables constructed from computationally cheaper quantities that remain strongly correlated with more expensive target observables. We can utilise surrogate variables (provided a strong systematic correlation across the ensemble), combined with a bias-reduction method to mitigate biases caused by their generation. Traditional surrogate variables take approaches such as the truncated solver method (which inspires the bias correction approach this study explores) and ratio methods, which rely on variance reduction. To compute the bias correction using the truncated solver method, we need to utilise a subset of the data that contains actual values and their surrogate estimates. From this subset, we find the expected difference between the values for an observable (O_{true}) and predicted expectation values (O_{true}), which we can store. We can then use this difference on uncorrected surrogate variables for a more unbiased set of variables. This can be shown in the formula below:

$$\langle O\rangle_{cor} = \langle O_{pred} - O_{true}\rangle_{bc} \quad (6)$$

Where $\langle O\rangle_{cor}$ is the correction vector for the observable O. The expectation vector is computed on the bias correction (bc) subset of data. Which is then applied via the following equation:

$$\langle O_{pred}\rangle_{corrected} = \langle O_{pred}\rangle_{uc} - \langle O_{cor}\rangle_{bc} \quad (7)$$

Where $\langle O_{pred}\rangle_{uc}$ is the uncorrected predictions of surrogate variables. $\langle O_{pred}\rangle_{corrected}$ is the corrected predictions and $\langle O_{cor}\rangle_{bc}$ is the bias correction vector in Equation 5 above. Lawrence (2025)

Meanwhile, for traditional variance reduction methods such as the ratio method, we must find the correction factor from 2 observables (O_1 and O_2) for which we have high precision data and low precision data (predictions) for O_1 and low precision data (predictions) for O_2. With this, we can try to predict a higher precision variant of O_2. This method tends to be a very good approximation towards the true values (which is why using Machine Learning models with Ratio method corrections is used as a benchmark in this paper). The method for applying is shown below:

$$O_{2,HP} = O_{1,HP}\frac{O_{2,LP}}{O_{1,LP}} \quad (8)$$

Where HP represents high precision and LP represents low precision. El-Khadra & Vega (2024)

2.6 Machine Learning in Lattice QCD

Machine Learning in Lattice QCD is an extension of traditional surrogate variable methods, providing more nonlinear mappings between correlated observables. This increases the flexibility in which we can construct surrogate variables and allows us to utilise relatively computationally cheap correlators (such as the Goldstone Operator D mesons) functions to predict more computationally expensive correlators (akin to the non-Goldstone operator D mesons). Tomiya (2025)

3 DATA AND METHODOLOGY

3.1 Aims

The goal of this experiment was to investigate bias-reduction factors and their use in correcting learned biases in generating surrogate variables via Machine Learning. For the experiment, we prioritised machine learning models using regression methods to learn the mapping between D mesons generated from Goldstone Operators and those generated from non-Goldstone operators and investigated their effects on truth data. We also used bias reduction methods to determine which method provided the best approximation to the truth data. The mapping is shown below:

$$D_{gold} \rightarrow D_{nongold} \quad (9)$$

3.2 Data

The dataset was provided as a raw, unbinned gpl (tab-separated) file consisting of 2-point and 3-point correlator functions (each appropriately labelled), with each row representing a lattice gauge configuration. The correlators present included D mesons and K mesons. This study explicitly focuses on D mesons constructed from Goldstone and non-Goldstone operators. K mesons were built with different momentum translations ($qmax$, $\frac{2qmax}{3}$, $\frac{qmax}{3}$, $q0$). The time extent of the lattice is $T = 96$; the exact spatial dimensions were not provided when the dataset was sourced. For each gauge configuration correlators were measured via 4 different source time slices. This dataset has a fine lattice spacing of $a = 0.06fm$. The raw dataset layout is provided below:

[tag containing correlator information].ll C(1) C(2) . . . C(96)

For example:

2pt _D _Gold _fine.ll C(1) C(2) . . . C(96)

For the D_{gold} dataset a total of 553 configurations (each with 4 source time slices) were present (a total of 2212 rows of data). For $D_{nongold}$ a total of 619 configurations were present (each with 4 source time slices). This in turn meant we had to utilise only 553 gauge configurations to ensure a 1-to-1 relation between D_{gold} and $D_{nongold}$ configurations.

3.3 Preprocessing

For this experiment, the data were extracted from the original gpl file and sorted into CSV files by correlator type, meson, and operator or momentum translation. This ensured that all D_{gold} data and $D_{nongold}$ remained separate.

To ensure accurate results, a separate process was run to average the correlator data across all 4 time sources, then binned with a bin size of 10 (to prevent overfitting in noisy regions). Jackknife means, jackknife samples and jackknife errors were generated in this averaging (which were to be used in fitting).

For machine learning, each row of the unaveraged correlator data (including those of different source time slices but same gauge configuration) from the set of D_{gold} and $D_{nongold}$ was treated as an independent gauge configuration (to ensure the models have more samples to learn on and prevent underfitting and mitigate significant biases from one configuration to the other.

The machine learning data was then transformed by a natural logarithm scale, resulting in:

$$C(t) \rightarrow ln(C(t)) \tag{10}$$

The log-transformed data was scaled using scikit-learn's StandardScaler (with mean = 0 and standard deviation = 1) to mitigate the significant impact of noisier regions.

3.4 Machine Learning

The scaled and log-transformed data is then split by the following standards (following closely to the approach taken by El-Khadra & Vega (2024)). The splits of the data are shown below:

Dataset	Split Percentage
Training	15%
Validation	15%
Bias Correction	15%
Testing	55%

After the data was split, model training could commence with D_{gold} data as inputs and $D_{nongold}$ data as targets. In this experiment, four different machine learning models were trained. The first of which was a Gradient Boosted Regressor Tree, which combines multiple weak decision trees (each learning an aspect of the data) into a larger, more accurate model. Following shortly was the Multi-Layer Perceptron, which utilises layers of neurons to form a neural network that effectively predicts the data. Finally, the last two models trained were the 1-Dimensional Convolutional Neural Network and the Regression Encoder Transformer.

Once the models are trained, a set of predictions are obtained from unlabelled data which are then inverse transformed and scaled up by inverting the log-transform.

3.5 Bias Reduction

The models are then used to generate bias reduction factors for the bias correction method (equation 7) and ratio method (equation 8) on the bias correction dataset. These reduction factors are then applied to the test data to create bias-corrected and ratio-method-corrected versions of the data.

3.6 Analysis

All the data obtained from the truth, machine learning models and the bias reduction factors are then analysed in two parts. A preliminary analysis is made where the effective mass of the dataset is determined via the formula shown in Equation 5.

We can then utilise a sliding window method (with a window size of 4) to see where the effective mass remains relatively stable and assign that as a prior value for our ground state effective mass (which, when using natural units, corresponds to our E_0) to be used by our fitting shortly after. Similarly, we can make approximations for our a_0 value from our extracted E_0 values, assigning relatively large widths to a_0 while maintaining a lower width for E_0 later during fitting. Lepage et al. (2001)

We now generate bins for our bias-reduced (both the ratio method and the bias correction method) and raw machine learning correlator data of size 10 and generate jackknife samples as well.

3.7 Fitting

Utilising the python package gvar (Lepage (2024a)) to create a dataset of the binned jackknife samples for each set of data (truth, raw ML, ML+BC, ML+RM) and a model through corrfitter (Lepage (2024b)) consisting of a two state fitting model (as shown in Equation 4) with a t_{min} = 10 (to reduce effect from higher energy excited states and stabilise the fit). Using the $ln(a_0)$ and $ln(E_0)$ as priors, which are obtained from the analysis step, we can try and fit the data effectively using lsqfit (Lepage (2024c)) to get a bayesian fit for the data, a fitted estimate of ground effective mass E_0 (also written as dE_0 in tables) and ground state adequate amplitude a_0. We can also obtain approximations for the first excited state effective amplitude a_1 and first excited state effective mass E_1 as well as the difference between ground and first excited state masses (dE_1).

4 RESULTS

4.1 Truth vs raw Machine Learning correlators

Figure 1 shows the performance of each of the 4 ML models against the Truth data for $D_{nongold}$ correlators. Models tend to capture the noisy nature of non-Goldstone operators well; however, they often overfit to the noise, leading to reduced fit accuracy. There is a reduction in accuracy as time approaches the temporal inflection point (T = 48).

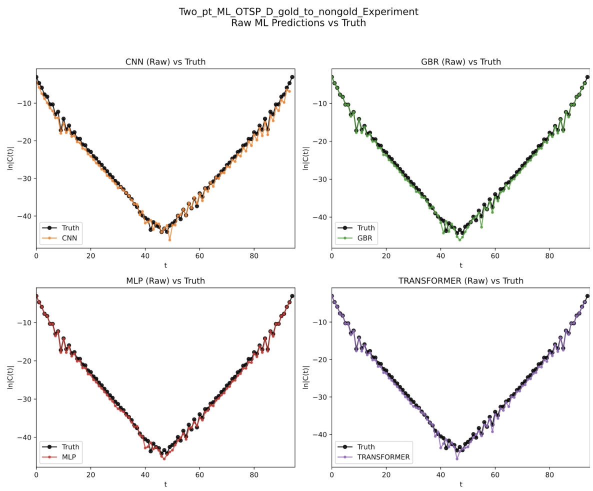

Figure 1. $ln(C(t))$ vs t (Log Correlator vs Euclidean time) for Raw Machine Learning and Truth (time source averaged) Correlators

4.2 Truth vs Bias Corrected correlators

Figure 2 displays the performance of each of the 4 ML models, post-bias correction, against the truth data for $D_{nongold}$ correlators. The bias correction clearly shows an improvement in the signal towards the inflection point; however, it tends to translate the earlier points slightly above the actual values.

4.3 Truth vs Ratio Corrected correlators

Figure 3 shows the performance of each of the 4 ML models, post-ratio method correction, against the truth data $D_{nongold}$ correlators. The ratio method performs remarkably well for earlier time points; however, it performs poorly near the point of inflection, faring even worse than the raw ML correlator data.

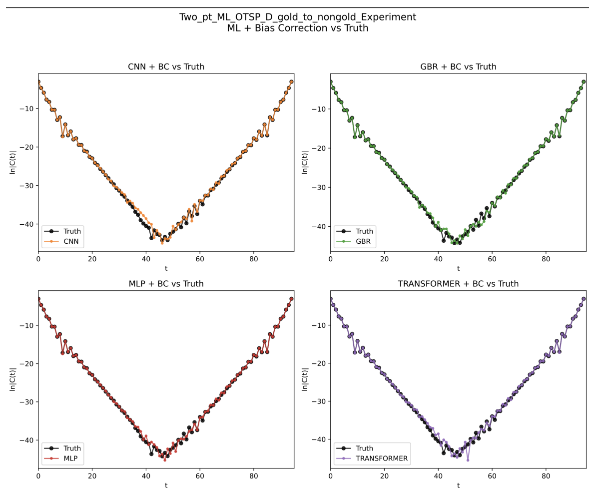

Figure 2. $ln(C(t))$ vs t (Log Correlator vs Euclidean time) for Bias Corrected Machine Learning and Truth (time source averaged) Correlators

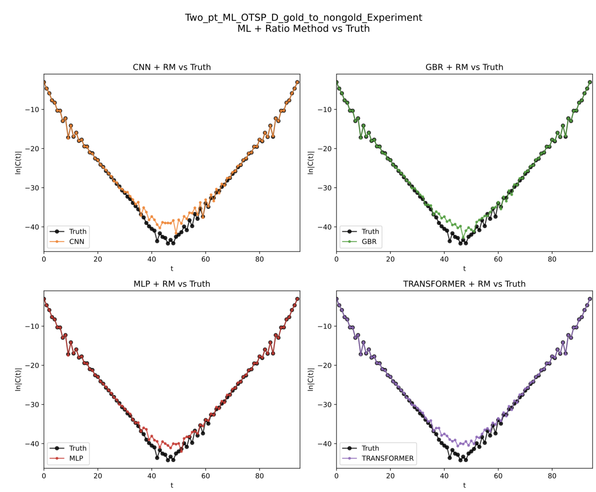

Figure 3. $ln(C(t))$ vs t (Log Correlator vs Euclidean time) for Ratio Method Corrected Machine Learning and Truth (time source averaged) Correlators

4.4 Noise-to-Signal ratios

Figure 4 shows the noise-to-signal against time ratio plot for all machine learning models and the truth data. The data towards the point of inflection is many orders of magnitude noisier than the data towards the tail ends of the correlator function (lowest and highest times). This implies both a weak signal and a noisier output at the point, as explained in the Theory section 2.4.

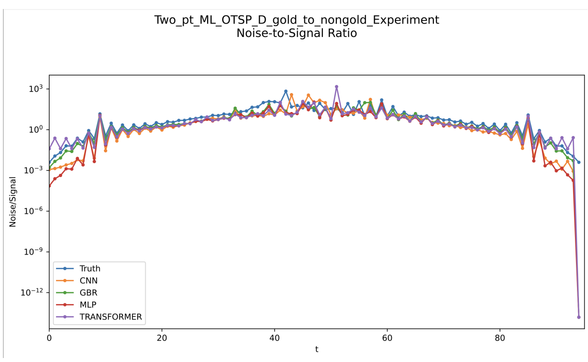

Figure 4. Noise-to-Signal ratio vs t for truth data as well as all ML models

4.5 Fits

: Figure 5 is a table that shows a direct comparison of the truth data, bias-corrected data for each ML model, and ratio-corrected data for each model. The table shows and compares the ground-state effective mass (dE_0), the ground-state effective amplitude (a_0), the difference between the ground and first excited states' effective mass (dE_1), and the first excited-state effective amplitude (a_1). All fits have a $\chi^2/d.o.f$ ranging between 0.5 and 1.2 making the fits very good estimates for the true correlator data.

Model	Method	dE0	dE1	a0	a1	χ²/dof	Q
Truth	-	0.8412(43)	0.1999(554)	0.0480(14)	0.0033(40)	0.802	0.755
CNN	BIAS	0.8315(87)	0.2000(416)	0.0434(42)	0.0034(30)	0.524	0.954
CNN	RATIO	0.8172(221)	0.2000(832)	0.0375(101)	0.0049(88)	0.963	0.497
GBR	BIAS	0.8329(73)	0.1999(554)	0.0437(33)	0.0033(40)	1.227	0.215
GBR	RATIO	0.8268(89)	0.1998(554)	0.0409(39)	0.0063(76)	1.343	0.135
MLP	BIAS	0.8315(104)	0.1999(1386)	0.0431(56)	0.0033(100)	1.461	0.099
MLP	RATIO	0.7798(456)	0.2000(832)	0.0225(155)	0.0060(107)	0.939	0.501
TRANSFORMER	BIAS	0.8216(115)	0.2000(693)	0.0384(55)	0.0033(50)	0.840	0.648
TRANSFORMER	RATIO	0.8097(195)	0.2000(693)	0.0344(83)	0.0062(94)	0.878	0.601

Figure 5. Table showing Correlator fit data at t_{min} = 12 for Truth data, Bias corrected model data and ratio method corrected model data

5 DISCUSSION

5.1 Machine learning for surrogate observables

The results explicitly show that supervised machine learning models can actively learn mappings between Goldstone and non-Goldstone operators across a lattice from the resultant gauge configurations. Machine Learning models tend to perform very effectively in mapping between these 2 correlators for the D mesons white also maintaining relatively stable outputs. However, due to the statistically noisy nature of the configurations towards the point of inflection the machine learning models do not perform at their best over here. They still remain a viable approach towards fitting correlator data even without any major corrections. As such they are a very good modern alternative to generating surrogate observables from lattice data once you have stable mappings. However, due to the nature of regression tasks (and machine learning in general) they require at least one large set of data to train on initially, to effectively learn mappings.

5.2 Bias correction vs Ratio Method

As seen from figure 2 and figure 3, we see that the ratio method tends to perform very effectively at times further from the point of inflection, at these points higher energy states are more prevalent which illustrates at the effectiveness of the ratio method at being more useful in situations were knowledge of the higher excited state masses are important (which is an area in which the bias correction method lacks). However, the ratio method does suffer from (much like as mentioned in the introduction) its multiplicative methodology as much as it benefits from it; this is evident in the stark difference between the truth values and the ratio method values (which are often far bigger than the truth), meaning it tends to be more sensitive to noise at later times. The ratio method results in consistent fits throughout all models despite its drawbacks however its predictions for the effective mass are often out of the range of error of the truth.

The bias correction method is slightly different in its approach, as we see that it tends to perform less effectively against the truth, as the ratio method did for earlier times. This is due to its additive nature, leading to overtranslation at earlier times. The method tends to be more impressive towards the point of inflection, where the ground state mass tends to be more prevalent, and the statistical noise factors dominate. Due to its additive nature, it's not as easily affected by reducing denominator terms like the ratio method and tends to improve the raw ML correlators at these points. The method does not have the best fit for the 1D CNN compared to the truth and ratio methods, but tends to perform better than the ratio method for the GBR and MLP models and tends to have effective mass values closer to the effective mass from the truth.

5.3 Limitations

This study is heavily limited by the size of the available dataset and the irregular number of configurations for D_{gold} and $D_{nongold}$, which impacts model training and loss reduction. As the dataset is limited, we lack additional data from different configurations to monitor overfitting or assess generalisation. This is critical because deep learning architectures tend to overfit datasets easily. As well as this, for models that tend to focus on learning both the content and structure of the correlator data (such as the 1D CNN and Encoder Regression Transformer) rather than mapping via a set of functions (like GBR and MLP do) we often see issues in them not being able to fit as effectively at earlier times due to their inability to recognise data from earlier, higher energy excited states.

Finally, the correction methods allow us to have only one type of desired information. If we wish to be more accurate for higher-energy states, we need to utilise ratio methods; meanwhile, for ground states, bias correction is more effective, resulting in a trade-off based on the experiment.

5.4 Further Improvements

To improve this study further, we can utilise datasets containing a larger quantity of D_{gold} and $D_{nongold}$ data and compare the effectiveness of models pretrained on this in a larger dataset to test generalisation ability and also see the effectiveness of newer models trained on a larger set of data.

We can also implement different loss functions that place greater weight on the earlier sections of the data to ensure the sequential models learn the excited-state areas better.

Instead of using a standard CNN architecture, which tends to pad at the ends of the data, we can utilise circular CNNs to prevent this padding from distorting data at the tail ends and compare the effectiveness of both models.

We can also implement machine-learning-specific bias-reduction methods that employ decaying multiplicative factors on top of a bias-corrected factor, weighted to be lower at earlier times and more relevant near the point of inflexion. This allows us to maintain both the additive and multiplicative properties of these reduction factors and develop a more accurate constrained Bayesian curve fit for the correlators.

6 CONCLUSION

This project explored utilising supervised machine learning with traditional surrogate-observable creation methods to generate observables for lattice QCD correlators, in an effort to reduce the computational cost of Dirac Operator inversion. The bias reduction methods tested in this study worked in tandem with the raw correlators generated by the machine learning models to produce correlator data that closely matched the truth data. While the ratio method showed excellent fit for earlier times and often reduced in performance for times closer to the point of inflection, the bias correction method yielded a broader set of results than the ratio method. However, it was less useful for earlier times. Both produced very reliable fits, often in close tandem with the truth fits when constrained curve-fitting methods were applied. With further refinements to model architectures and physics-informed losses, we can ensure even more accurate surrogate observable generation. At the same time, more data can help us test how well our models generalise. Additionally, a bias reduction function that combines the ratio method at earlier times and bias correction methods at later times (closer to the point of inflection) can generate better observables even with traditional methods.

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DATA AVAILABILITY

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7 NOTES

This template is very opinionated on how it prefers to utilise images as such there is a rather inappropriate amount of whitespace within sections 4.1 to 4.5 which I cannot format around.

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